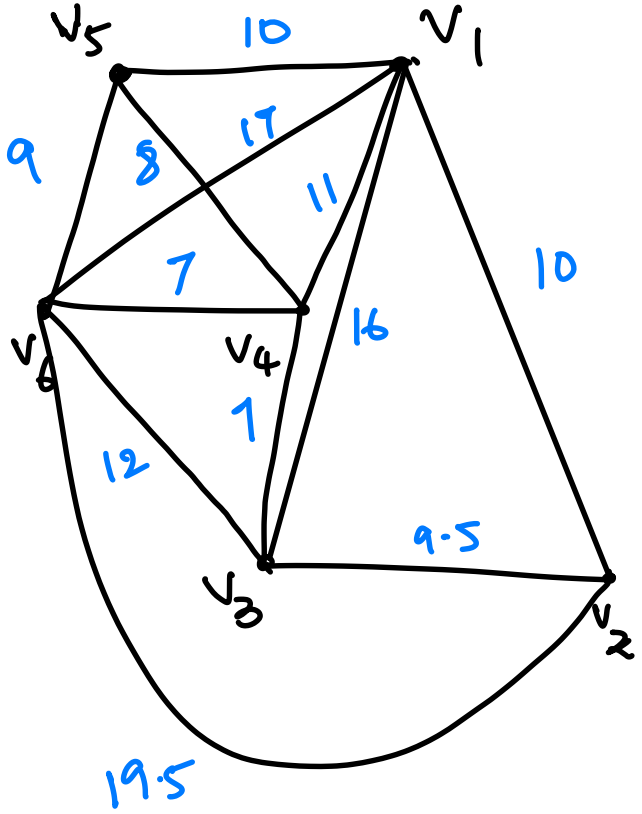
height of tree $\in \Omega(n \log n)$

$\therefore$ Prim's algorithm takes $\Omega(n \log n)$ time

Note:

Data structure in Prim's & Dijkstra's

is Heap.


v_5
10
v_1
9
8
17
11
7
10
v_6
v_4
16
7
12
9.5
v_3
v_2
19.5

(Last Class eg)

1. V_1V_2 – 10

2. V_2V_3 – 9.5

3. V_3V_4 – 7 ✓

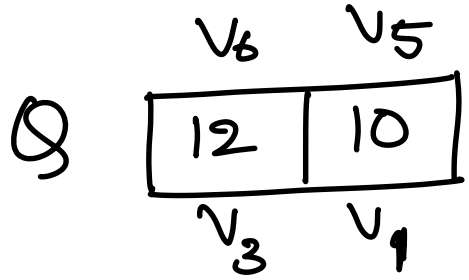
4. V_4V_6 – 7

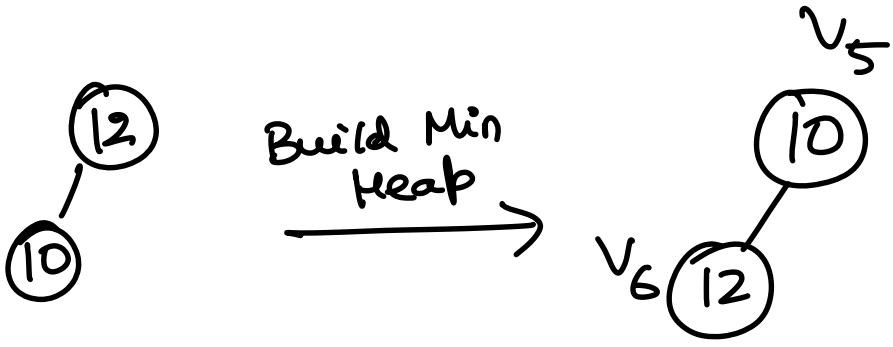
5. V_4V_5 – 8

41.5

Extract Min Q.

min ← 7



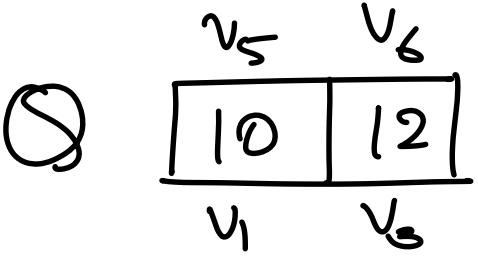


Min Heafify

i = 1

12 ≮ 10 10 ⇔ 12

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$u \leftarrow v_4$

$Adj[v_4] = \{v_1, v_5, v_6, v_3\}$ (11, 8, 7, 7)

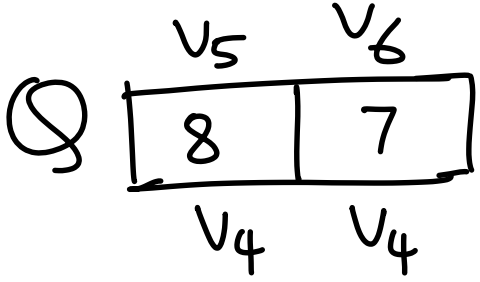
$v_1 \notin Q$

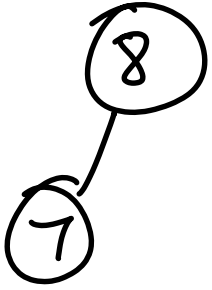
$v_5 \in Q$ and $w(v_4, v_5) = 8 < 10$ update

$v_6 \in Q$ and $w(v_4, v_6) = 7 < 12$ update

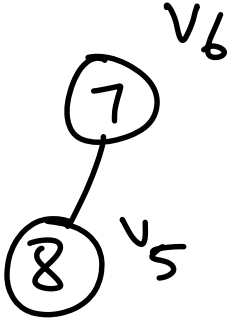
$v_3 \notin Q$

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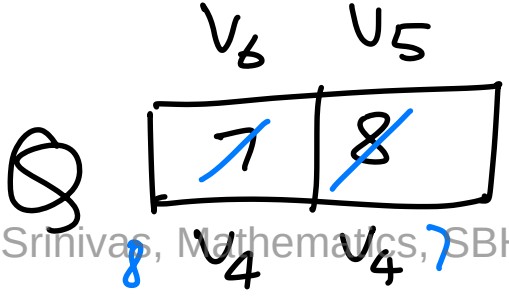
Build Min heap →



$i=1$

$8 \nleq 7$

$7 \leftrightarrow 8$.



Extract Min Q.

min ← 7



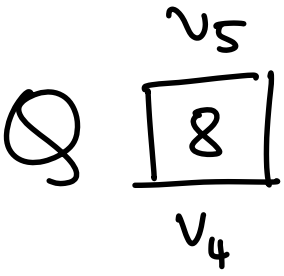
$u \leftarrow v_6$

$Adj[v_6] = \{v_1, v_2, v_3, v_4, v_5\}$ (17, 19.5, 12, 7, 9)

$v_1, v_2, v_3, v_4 \notin Q$, $v_5 \in Q$ $\quad w(v_6, v_5) = 9 \not< 8$

No change

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Valid Heap

Extract Min Q.

$min \leftarrow 8$.

$Q = \phi$.

$u \leftarrow v_5$

$Adj[v_5] = \{ v_1, v_4, v_6 \}$ (10, 8, 9)

$v_1 \notin Q$, $v_4 \notin Q$, $v_6 \notin Q$

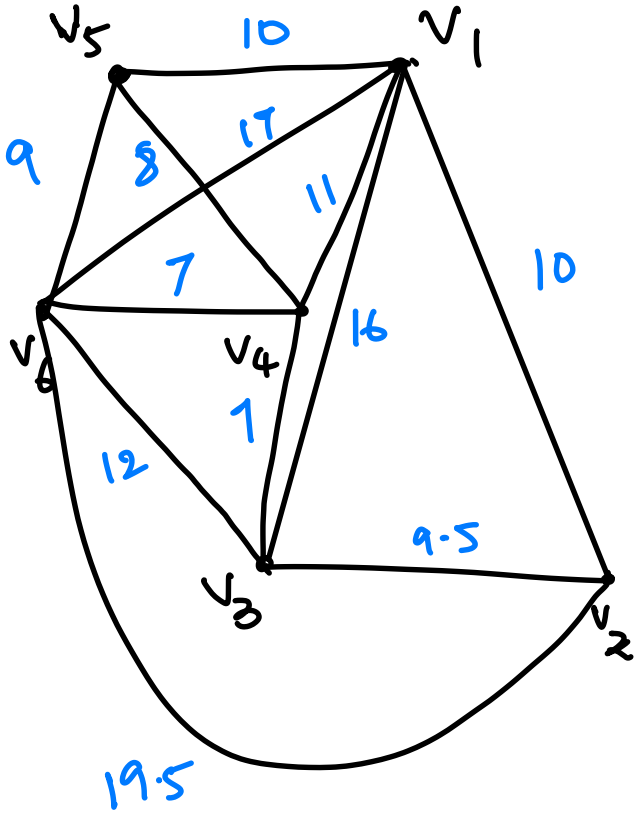
$Q = \phi$ (Stop)

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Kruskal's Algorithm :

1. Sort the edges by weight
2. Build a spanning forest by adding the edge of minimum weight which when added to those already chosen does not form a cycle.

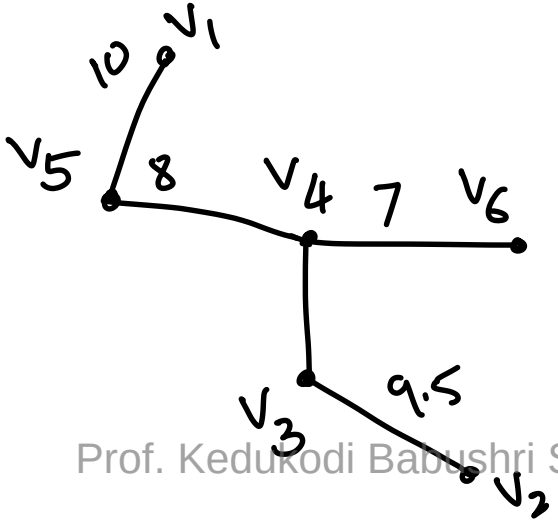
eg:



1. Sort the edges.

$\because v_5v_6$ forms a cycle.

v_3v_4	v_1v_6	v_4v_5	~~v_5v_6~~	v_3v_2	v_5v_1	v_1v_2	v_4v_1	v_3v_6	v_3v_4	v_6v_1	v_6v_2
7	7	8	9	9.5	10	10	11	12	16	17	19.5



Choose if does not form cycle

1. v_4v_6 – 7
2. v_3v_4 – 7
3. v_4v_5 – 8
4. v_3v_2 – 9.5
5. v_1v_5 – 10

41.5

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* Implementation of 'Kruskal's algorithm involves a data structure called

Disjoint - Set Forests.

Operations Involved :

Make-Set(x)

1. $P[x] \leftarrow x$
2. $rank[x] \leftarrow 0$

$P[x] \rightarrow$ parent of node x

$rank[x] \rightarrow$ no. of edges in the longest path between x and a leaf.

Union(x, y).

1. Link (Find-set(x), Find-set(y))

Link(x, y).

1. If rank[x] > rank[y]
2. then P[y] ← x
3. else P[x] ← y
4. If rank[x] = rank[y]

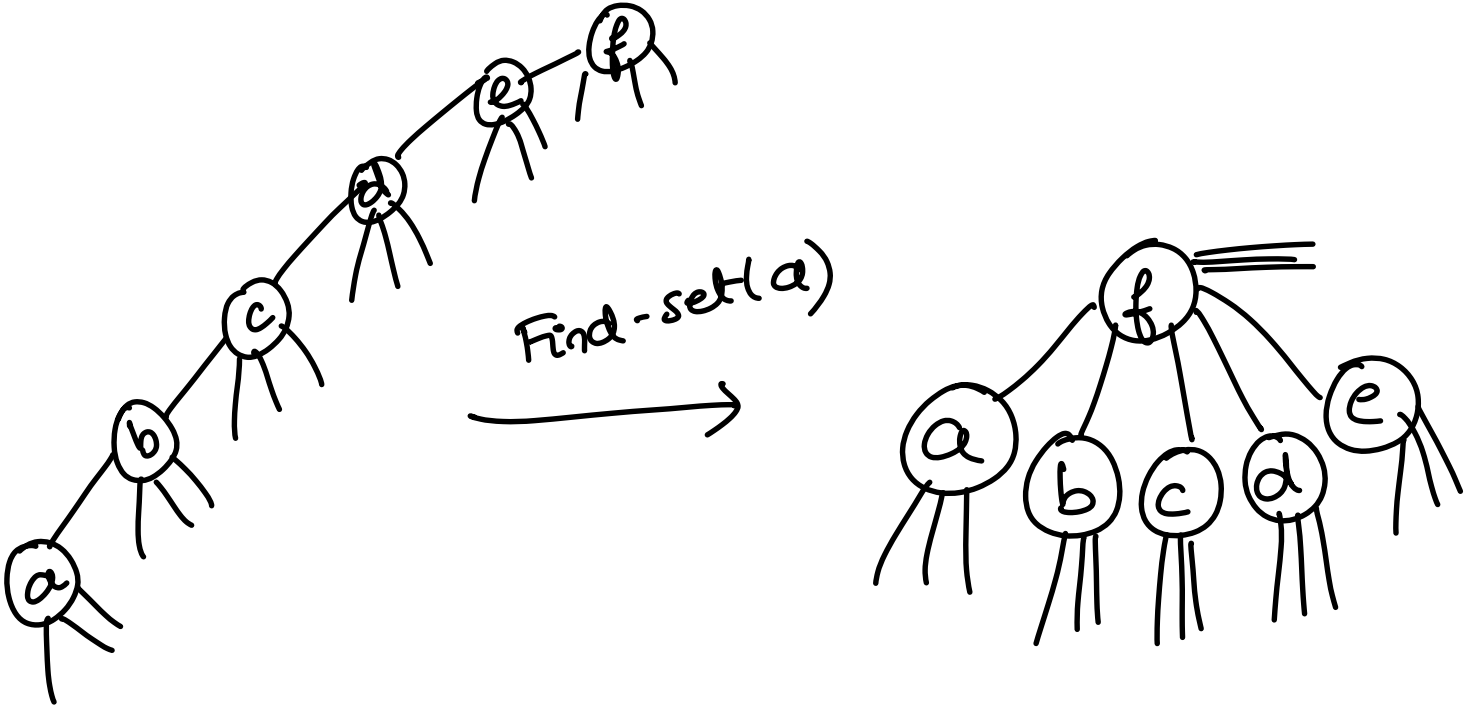
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5. then rank[y] ← rank[y] + 1.

Find-set (x)

1. If $x \neq P[x]$
2. then $P[x] \leftarrow$ Find-set $(P[x])$
3. return $P[x]$

Implement Find-set(x) on :



Find-set(a)

$P(a) = b \neq a$

$P[a] \leftarrow$ Find-set $(P(a))$

$=$ Find set (b)

$P(b) = c \neq a$

$P(b) \leftarrow$ Find-set (c)

Find set(c)

$c \neq P(c) = d$

$P(c) \leftarrow$ Find set(d)

Find set(d)

$d \neq P(d) = e$

$P(d) \leftarrow$ Find set(e)

$e \neq P(e) = f$

$P(f) \leftarrow$ Findset(f)

$P(f) = f$

returns $P(f) = f$

$\Rightarrow P(e) = f$

$\Rightarrow P(d) = f$

$\vdots$

$\Rightarrow P(a) = f$

MST-Kruskal (G, w)

1. $A \leftarrow \phi$
2. For each vertex $v \in V(G)$
3. do Make-Set (v)
4. Sort the edges of graph G into non-decreasing order by wt w.
5. For each edge $(u, v) \in E$ taken in non-decreasing order by wt

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6. do if Find_set(u) ≠ Find set(v)

7. then A ← A ∪ {(u, v)}

8. Union(u, v)

9. return A.

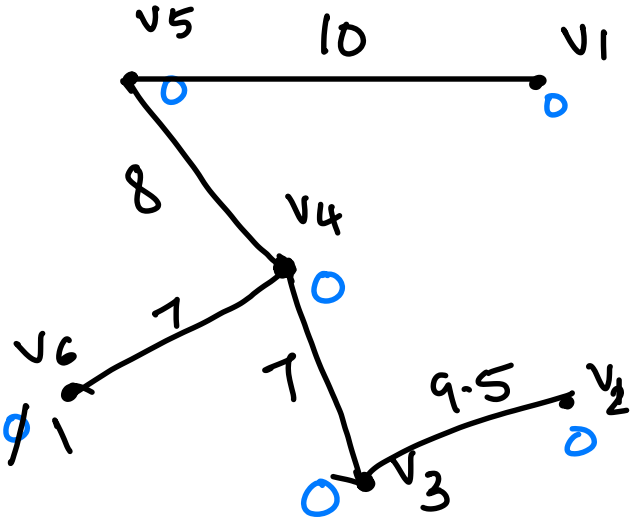
Operations Involved:

Make-Set (x)

1. $P[x] \leftarrow x$
2. $rank[x] \leftarrow 0$

$P[x] \leftarrow$ parent of node x

Each vertex parent is itself and rank 0



Sorted :

all parent = v_6 = root.

E

✓	✓	✓	X	✓	✓	X	X	X		
~~v_4v_6~~	v_3v_4	v_4v_5	v_5v_6	v_2v_3	v_5v_1	v_1v_2	v_4v_1	v_3v_6	v_3v_1	v_6v_1	v_6v_2
7	7	8	9	9.5	10	10	11	12	16	17	19.5

$(u,v) = (v_4, v_6) \in E$ ✓

Find-set(v_4) = v_4 ($\because P[v_4] = v_4$)

Find-set(v_6) = v_6 ($\because P[v_6] = v_6$)

$v_4 \neq v_6$

$A \leftarrow \phi \cup \{(v_4, v_6)\} = \{(v_4, v_6)\}$

Union (v_4, v_6)

$\|$

Link(Find-Set(v_4), Find-Set(v_6))

$=$ Link(v_4, v_6)

ie add an edge and increase rank or update parent.

Did p[v_4] update?

$0 = \text{rank}[v_4] = \text{rank}[v_6] = 0$

$\Rightarrow \text{rank}[v_6] = 0 + 1 = 1$

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and add the edge $v_4 v_6$.

$(u,v) = (v_3, v_4) \in E$

Find_set $(v_4) = v_6 \neq v_3 =$ Find_set (v_3)

$A \leftarrow A \cup \{(v_3, v_4)\}$

$= \{(v_4, v_6), (v_3, v_4)\}$

Union (v_3, v_4)

↓

Link(Find-set(v_3), Find_set(v_4))

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